GGG-H-177B
February 1, 1972
SUPERSEDING
Fed. Spec. GGG-H-177a
October 16, 1964

FEDERAL SPECIFICATION

HELMET, SAFETY, ELECTRICAL WORKER'S

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers insulating safety helmets for use of linemen and other electrical workers as protection against high voltage electric shock and burn, and against falling or flying objects.

1.2 Classification.

1.2.1 Types. The helmets shall be furnished in the following types:

Type I - Full brim (hat).
Type II - Brimless with visor (cap).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-636 - Box, Fiberboard.
PPP-F-320 - Fiberboard; Corrugated and Solid Sheet Stock (Container Grade), and Cut Shapes.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 406 - Plastic: Methods of Testing.
Fed. Std. No. 595 - Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following publications of the issues in effect on date of invitation for bids form a part of this specification.

American National Standards Institute, Inc.

American National Standard Safety Requirements for Industrial Protective Helmets for Electrical Workers, Class B.ANSI.Z89.2-1971.

(Application for copies should be addressed to American National Standards Institute, Inc., 1430 Broadway, New York, New York 10018.)

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the National Motor Freight Association, Inc., Agent 1616 P Street N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Qualification. Safety helmets furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein (see 6.2).

3.2 Component parts. Each helmet shall consist of a shell, headband, sweatband, crown straps forming a suspension for supporting the helmet on the wearer's head and, if specified, (see 6.1) an adjustable chin strap and a winter lining.

3.3 Materials. All materials used in the construction of the helmets shall conform to the requirements for class "B" helmets of ANSI.Z89.2-1971, except as hereinafter specified.

3.4 Detail construction requirements.

3.4.1 Shell. The shell shall be dome shaped, without holes, and of one piece seamless construction with a full brim for type I helmets and brimless with a visor for type II helmets. The area under the peak or front of the helmet brim may be covered with a nonconducting anti-glare material.

3.4.1.1 Distinguishing identification markers on the inside or outside of the shell surface shall be of nonconducting material and shall be moulded, cemented, or otherwise affixed to the surface without the use of rivets, screw holes, or indentations of any nature. Paint or ink of a permanent nature may be used for such markings and shall be of nonconducting material.

3.4.1.2 Color. Unless otherwise specified the shell of the helmet shall be yellow approximately matching color No. 23793 of table V of Fed. Std. No. 595 except the finish shall be gloss in lieu of semi-gloss. When paints are used for coloring the shell caution shall be used by the manufacturer inasmuch as some paints reduce dielectric protection, attack and soften the shell material.

3.4.2 Brim.

3.4.2.1 Type I, full brim. The brim for type I helmets shall be an integral extension of the shell dome and shall be continuous around the helmet. The width of the brim shall be not less than 1-1/4 inches.

3.4.2.2 Type II, brimless with visor. Type II helmets shall be brimless with a visor (peak) extending from the front of the crown not less than 1-1/2 inches and shall be not less than 5-1/2 inches wide at the narrowest width.

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3.4.3 Headbands. The headbands shall be adjustable, and shall permit adjustment to all sizes within the range permanently marked on the headbands. The headbands, which shall measure a nominal width of 1 inch and which may be a replaceable component, shall be securely attached either to the helmet shell or to the crown straps for assuring its proper position in the helmet. Attachments shall be so constructed that there will be no sharp edge, burrs or protrusions that will cause injury to the user on impact, and the manner of the attachments shall not interfere with the adjustments of the crown straps. Size markings shall be marked on headbands and shall range from 6-1/2 to 8 inclusive and shall be adjustable in 1/8 size increments. The clearance between the inside of the shell and the headband shall be not less than 1/4 inch or more than 3/4 inch. Materials shall be non-toxic and nonirritating.

3.4.4 Sizes (for headbands). Helmets shall have the following adjustable headband sizes and circumference measurements as specified in table I.

TABLE I. Sizes

Sizes (headband)	Circumference (inches)
6-1/2	20-1/2
6-5/8	20-7/8
6-3/4	21-1/4
6-7/8	21-5/8
7	22
7-1/8	22-3/8
7-1/4	22-3/4
7-3/8	23-1/8
7-1/2	23-1/2
7-5/8	23-7/8
7-3/4	23-1/4
7-7/8	24-5/8
8	25

NOTE: The above measurements are to be made with a measuring tape that will not stretch. In selecting sizes, measure the circumference of the head at the location where the helmet is normally worn. Note the nearest corresponding figure for size in the table. Allowable tolerance of circumferential measurement shall be plus or minus 1/8 inch. Nothing in this specification shall be construed as prohibiting larger or smaller headband sizes than specified. Headbands that incorporate an integral nape strap will not necessarily conform to the circumferential measurements tabulated above; however, they must accommodate the required head sizes. Additional headbands per helmet are acceptable for size range coverage.

3.4.5 Sweatbands. The sweatbands shall be made of material in accordance with approved materials for sweatbands under Z89-2-1971. Sweatbands shall be of the removable-replaceable type, or may be integral with the headband. The sweatbands shall cover not less than the forehead portion of the headband, and the sweatband assemblies shall contain no metal or electrical conducting materials or materials that are toxic or irritating.

3.4.6 Crown straps. The crown straps, when assembled, shall form a cradle for supporting the helmet on the wearer's head so that the distance between the top of the head and the underside of the shell cannot be adjusted to less than 1-1/4 inches as measured under test conditions in accordance with 4.6.3.3. The upper extremities of the strap assembly shall be securely attached by an all nylon cord or lace, or shall be a continuous length of one piece strap construction. Materials used shall be non-toxic and nonirritating.

3.4.7 Accessories. If chinstraps and winter liners are specified in the contract or order, they shall conform to the requirements of Section 5.5 of Z89.2-1971. Materials shall be non-toxic and non-irritating.

3.5 Physical requirements.

3.5.1 Leakage current and voltage resistance. Helmets, when tested in accordance with 4.6.2, shall withstand 20,000 volts, AC, 60 cycles per second (RMS Value) for 3 minutes with a leakage of not over 9 milliamperes. When tested to breakdown the helmet shall not fail below 30,000 volts. Each helmet tested shall first pass the mechnical proof test, 4.6.3.3.2.

3.5.2 Impact absorption. Helmets shall not transmit an average force of more than 850 pounds and no individual specimen shall transmit a force greater than 1,000 pounds when tested in accordance with 4.6.3.3.1. When tested in accordance with the mechanical proof test, 4.6.3.3.2, helmets shall show no evidence of substantial contact between the shell, suspension, and headform.

3.5.3 Penetration strength. The helmet shall not be pierced for more than 3/8 inch, including the thickness of the shell material, when tested in accordance with 4.6.4.

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3.5.4 Flammability. The thinnest section of the shell shall not burn at a rate greater than 3 inches per minute when tested in accordance with 4.6.5.

3.5.5 Water absorption. The helmet shell material shall not absorb more than 0.5 percent of water when tested in accordance with 4.6.6.

3.5.6 Weight. The weight of each complete helmet including the suspension and headband, but without accessories shall not exceed 15-1/2 ounces when tested in accordance with 4.6.9.

3.6 Interchangeability. The headband or sweatband, or both, and crown strap assembly shall be easily removable and replaceable, and shall be interchangeable among shells of the same size furnished by the same manufacturer (see 6.1.1).

3.7 Instruction sheet. An instruction sheet which explains the proper method of wearing and adjusting the helmet shall be furnished with each helmet. The sheet shall be stowed in the helmet so as to remain where placed until intentionally removed.

3.8 Toxicity. The qualifying agency shall determine that the finished item, when used for its intended purpose, is not likely to result in adverse effects on health of personnel. In the case of Army procurement, the Surgeon General of the Army will act as advisor to the procuring agency in making the determination. The supplier shall submit a certificate of compliance stating materials are non-toxic.

3.9 Marking.

3.9.1 Shell. Each helmet shell shall be marked in a permanently legibly manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined. In addition, each shell shall be marked to indicate that it meets the voltage test and breakdown requirements herein. When specified, the shell of each helmet shall be permanently marked with "U.S. Government".

3.9.2 Headbands. The head size and range of sizes to which the headband is adjustable shall be marked on labels cemented to, or embossed upon, the helmet shell and sewn or comented to the replaceable crown-strap assembly (see 3.4.4).

3.10 Workmanship. The helmets shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels (AQL's).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Qualification tests. Qualification tests shall consist of visual and dimensional examination and the applicable tests specified in 4.6. Qualification tests shall be conducted at a commercial testing laboratory approved by the Standardization Division, Federal Supply Service, General Services Administration. Approval of the laboratory for qualification testing shall be obtained from the Standardization Division prior to testing. Application for approval shall include the following information:

Name and location of laboratory.
Extent of laboratory's experience in testing helmets.
Statement that laboratory is equipped to make the applicable tests specified in 4.6.

4.2.1 Samples for qualification inspection. Eighteen helmets of each type and material for which approval is desired shall be furnished. Qualification will cover only those helmets of the material from which the sample was made.

4.3 Sampling for lot acceptance inspection.

4.3.1 Lot. All helmets of one type presented for delivery at one time shall be considered a lot for purposes of acceptance inspection.

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4.3.2 Sampling for visual and dimensional examination. A random sample of helmets shall be taken from each lot in accordance with MIL-STD-105 at inspection level III. The acceptable quality level (AQL) for major defects shall be 2.5 per hundred units (DHU) and for major and minor defects combine shall be 6.5 defects per hundred units (DHU).

4.3.3 Sampling for impact test and leakage current. A random sample of helmets shall be taken from each lot in accordance with MIL-STD-105 at inspection level S-4. Any departure from the requirements will be cause for rejection of the lot. The sample unit shall be 1 helmet. The leakage current and voltage resistance test shall be conducted after first subjecting the helmet to the impact test.

4.3.4 Sampling for penetration test. A separate random sample of helmets shall be taken from each lot in accordance with MIL-STD-105 at inspection level S-2. Any departure from the requirements will be cause for rejection of the lot.

4.4 Acceptance inspection.

4.4.1 Visual and dimensional examination. Each of the sample helmets taken in accordance with 4.3.2 shall be visually examined to verify compliance with table II. Examination shall be conducted as indicated below:

Any helmet in the sample containing one or more visual or dimensional major defects shall be rejected, and, if the number of defective helmets in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4.2 Acceptance tests. Samples taken in accordance with 4.3.3 and 4.3.4 shall be subjected to the tests specified in 4.6.2 through 4.6.9. If any sample helmet fails one or more of these tests, the entire lot represented by the sample shall be rejected.

4.4.3 Examination of preparation for delivery. An examination shall be made to determine whether packaging, packing, and marking requirements of section 5 are complied with. Defects shall be scored in accordance with the following: The sample unit shall be one shipping container fully prepared for delivery. Defects of closure listed in the subsequent table III shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-1 and the AQL shall be 4.0 defects per hundred units.

4.5 Rejected lots. Rejected lots may be offered again for inspection provided the contractor has removed all nonconforming helmets. Samples from such resubmitted lots shall again be taken and inspected to verify compliance with this specification.

TABLE II. Visual and dimensional examination

Examine	Defect	Major	Minor
Color	Color not in accordance with requirement.	X	
Design	Any characteristic not in accordance with specified requirements seriously affecting serviceability.	X	
	Any characteristic not in accordance with specified requirements affecting serviceability but not seriously.		X
Material (General)	Any component not fabricated of approved material seriously affecting serviceability.	X	
	Any component not fabricated of approved material not seriously affecting serviceability.		X
	Use of any metal components.	X	
Sweatband, headbands or other accessories (if applicable).	Side in contact with skin not smooth.		X
	Any deep cuts or holes seriously affecting serviceability.	X	
	Any cuts or holes affecting serviceability but not seriously.		X
	Spots or stains that are clearly noticeable.		X
	Hard or bony leather or excessively flanky leather seriously affecting serviceability.	X	

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TABLE II. Visual and dimensional examination (con.)

Examine	Defect	Major	Minor
Sweatbands, headbands or other accessories (if applicable). (con.)	Flanky leather affecting serviceability but not seriously.		X
	Any assembly that is not adjustable where required, e.g., chinstrap not adjustable.	X	
	Any sharp burr, sliver, or edge.		X
	Any component or assembly malformed seriously affecting serviceability.	X	
	Any component or assembly malformed affecting serviceability but not seriously.		X
	Any fastener, i.e., eyelets, rivets missing or not securely set to a degree where it may become detached from assembly.	X	
	More than one fastener not securely set but is not expected to become detached from assembly.		X
	Any fastener too tightly set or compressed resulting in damage to assembly, e.g., too much pressure applied to rivet resulting in damage to shell.	X	
Assembly highlights	Attachment of the cradle assembly to helmet does not provide adequate clearance between the shell and the headband.	X	
Crown straps and laces	Any rip, hole, cut or tear or other defect seriously affecting serviceability.	X	
	Defective to a degree where serviceability will be affected but not seriously, e.g., misweave, slubs, etc., if applicable.		X
Shell	Any hole, soft spot, air pocket, void, evidence of delamination, rough or abraded area, cracked, chipped, blistered, pitted or other defect seriously affecting serviceability.	X	
	Scratches, air pocket, void, rough or abraded area, chipped, pitted, blistered or other defect affecting appearance but not seriously affecting serviceability.		X
	Parts and clips for attaching crown straps and headbands to shell, positioned in manner likely to cause injury if blow on brim forces helmet over ears or forehead.	X	
Construction and workmanship general (applicable to all components, unless otherwise indicated herein).	Any component omitted, damaged, misplaced, operation omitted or not properly performed seriously affecting serviceability.	X	
	Any component misplaced, damaged, operation not properly performed affecting serviceability but not seriously.		X
	Any component not firmly affixed or attached to an extent that parts or assembly may become separated in use.	X	
	Any component not firmly affixed or attached but part or assembly is not expected to become detached in use.		X

TABLE III. Inspection of preparation of delivery

Examination	Defect
Marking (exterior and interior)	Omitted, incorrect, illegibly, of improper size, not in specified location, sequence, or method of application, incomplete.
Materials	Any component missing. Any component damaged, affecting serviceability.

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TABLE III. Inspection of preparation of delivery (con.)

Examination	Defect
Workmanship	Inadequate application of components, such as: Incomplete closure of case liners, container flaps, loose strapping, inadequate stapling. Bulging or distortion of containers.
Content	Number per container is more or less than required.
Weight	Gross weight exceeds requirements.

4.6 Procedures.

4.6.1 Preparation of samples. Where it is evident that the sample helmets have a protective coating over the basic material, the exterior surface of the shell is required to be abraded until the basic material is exposed, using number 60-grit garnet paper. Tests shall be made at 23° + 2°C. or 73.4° + 3.6°F. The temperature specified in test procedures shall be interpreted as the temperature of the specimen.

4.6.2 Leakage current test. The leakage current test shall be performed as follows to determine conformance with 3.5.1.

4.6.2.1 Apparatus. The test apparatus shall consist of the following:

(a) A vessel, of sufficient size, and containing fresh tap water, to submerge an inverted helmet to within 1/2 inch of the brim of the crown.
(b) Wire frame or non-conducting plastic frame for suspending inverted helmet in water.
(c) Source of 60 cycle alternating current of 30,000 volts (root mean square) maximum value.
(d) Wiring and terminals for application of voltage across dome of helmet.
(e) Voltmeter of sufficient capacity and accuracy.
(f) Milliammeter of sufficient capacity and accuracy to measure the specified current.

4.6.2.2 Procedure. Helmets shall be conditioned prior to this test by submerging them in tap water for a period of 24 hours, after which they shall be surface dried to prevent flashover during the test. Suitable precautions shall be taken to prevent accidental contact by persons with any part of the high-voltage circuit. An inverted helmet with the head harness assembly removed, shall be placed on the wire frame and submerged in the water to within 1/2 inch of the brim or visor edge as measured along the outside of the helmet brim or visor. The helmet shall also contain the water to the same level. One terminal from the electrical current source shall be connected to the wire frame which supports the helmet, and the other shall be immersed in the water contained in the helmet. Twenty thousand (20,000) volts shall be applied for three minutes and the current leakage noted. Voltage shall then be increased to 30,000 volts at a rate of approximately 1,000 volts per second, and held for three seconds, to determine whether breakdown of the shell occurs. At the end of the test period, the applied voltage shall be reduced to at least half-value before opening the test circuit.

4.6.2.3 Report. For each specimen, the leakage current or evidence of complete breakdown shall be reported.

4.6.3 Impact test. The impact shall be performed to determine conformance with 3.5.2.

4.6.3.1 Apparatus. The test apparatus shall consist of the following:

(a) The standard head forms mentioned throughout this specification are the model heads known as "A.M.L. Head Size Standards", medium size, a set of which is located at each of the following places:

National Bureau of Standards
Washington, DC 20234

Aero Medical Laboratory
Wright Air Development Center
Wright-Patterson Air Force Base, Ohio 45433

Nondestructive and Tool Testing Division
Laboratory Branch
Quality Assurance Section
Philadelphia Naval Shipyard
Philadelphia, Pennsylvania 19112

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NOTE: Test-block forms which simulate the standard headform may be employed. This headform may be low resonance magnesium K-1A, aluminum or wood. For protection from damaged wood, headforms may be provided with a steel insert in the crown.

(b) A Brinell penetrator assembly as shown in Figure 1. The impression bar shall be of 1100-0 aluminum 1/2 x 1-1/8 inch having a predetermined Brinell hardness of 21 to 24 as measured with a 500-kilogram load using a 10-millimeter ball. The Brinell penetrator used for the impact test shall be a hardened steel ball 1/2 inch in diameter.

(c) A steel ball approximately 3-3/4 inches in diameter and weighing 7.8 to 8.0 pounds. The ball shall be a smooth-surfaced steel sphere.

(d) A Brinell microscope or other suitable micrometer microscope accurate to 0.05 millimeter (mm.).

4.6.3.2 Test conditions.

4.6.3.2.1 For qualification tests. Impact tests shall be conducted with the specimens maintained at temperatures of $0^{\circ} \pm 2^{\circ}$F. and $120^{\circ} \pm 2^{\circ}$F. Specimens shall be subjected to the test temperature for at least 2 hours prior to the impact test.

4.6.3.2.2 For acceptance tests. Impact tests shall be conducted with the specimens maintained at the prevailing atmospheric temperature.

4.6.3.3 Test procedures.

4.6.3.3.1 Impact absorption test. For the impact absorption, mechanical proof, and penetration resistance tests, the specimen, with adjustment lace (if any) removed, and the headband adjusted to size 7-1/4, shall be mounted on the medium standard headform so that the drop ball, the headform and penetrator ball are center-aligned by means of a plub bob. For all low-temperature tests the headband shall be adjusted on the headform prior to conditioning by "opening" the headband the equivalent of two 1/8 head size adjustments. The center of the crown of the specimen shall be as nearly center as possible. The specimen shall be mounted with the back toward the fulcrum of the test equipment. To determine the clearance, the shell, minus suspension, shall be placed on the headform and a dimensional reading taken as shown in Figure 2. The suspension shall then be installed and another dimensional reading taken with a 25 pound weight, having a flat surface of not less than 2 inches in diameter, applied to the crown of the shell. The difference in dimensional readings shall be the clearance.

The Brinell penetrator assembly, with headform attached, shall be placed on a substantially level concrete floor and located beneath the drop ball so that the center of the headform is aligned with the center of the drop ball by means of a plumb bob. Specimens shall be tested at 0°F. and at 120°F. All specimens shall be subjected to the test temperature for at least two hours prior to impact tests. The impact test should be conducted within 15 seconds after the removal of the specimen from the temperature condition apparatus. The ball shall be dropped vertically on the crown from a height of 60 inches measured from the bottom of the ball to the top of the shell. The ball shall not be allowed to strike the specimen more than once. Impression shall be spaced at least 2-1/2 diameters apart, edge to edge, and not less than 2-1/2 diameters from the edge of the bar. Elliptical impressions shall be disregarded if the difference between the minimum and maximum axis exceeds 0.3 millimeter. All impressions from double blows shall be disregarded. The minimum diameter of the impression produced on the impression bar shall then be measured to the nearest 0.1 millimeter with the Brinell microscope. For each test the average Brinell hardness number of the impression bar used shall be recorded.

4.6.3.3.2 Mechanical proof test. The procedures as specified in 4.6.3.3.1 shall be followed, except:

(a) Helmets shall be tested at 0°F and 140°F.
(b) The Brinell penetrator assembly is not required.
(c) A sheet of carbon paper, 4 inches in diameter, shall be affixed to the underside of the top of the shell with the coated side downwards.

4.6.3.4 Reporting. The average force and greatest individual force for at least three specimens tested under one group shall be computed and reported. The forces shall be computed from the impression diameter using table IV or using the following Brinell formula:

$$F = 2.2 \times H \times \frac{\pi D}{2} \times \left(D - \sqrt{D^2 - d^2}\right)$$

Where:
"F" is the transmitted force in pounds,
"H" is the average Brinell hardness number of the impression bar,
"D" is the diameter of the impression ball in millimeters, and
"d" is the diameter of the impression is millimeters.
Values of "F" for various diameter of impression on impression bars of various hardness are tablulated in table IV.

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4.6.3.4.1 Whether or not there is evidence of substantial contact between the shell and the suspension or headform when conducting the mechnical proof test, the results shall be reported.

TABLE IV. Transmitted forces in pounds

Diameter of impression, millimeters	Brinell hardness number												
	18	19	20	21	22	23	24	25	26	27	28	29	30
3.9	485	515	540	565	595	620	650	675	700	730	755	785	810A
4.0	510	540	570	600	625	655	680	710	740	765	795	825	850
4.1	540	570	600	630	660	690	720	750	775	810	835	865	895
4.2	565	600	630	660	690	720	755	785	815	850	880	910	940
4.3	595	625	660	695	725	760	790	825	860	890	925	955	990B
4.4	625	660	690	725	760	795	830	865	900	935	970	1000	1040
4.5	650	690	725	760	795	835	870	905	940	975	1015	1050	1085
4.6	680	720	760	795	835	870	910	950	985	1025	1060	1100	1135
4.7	715	755	790	830	870	910	950	990	1030	1070	1100	1150	1190
4.8	745	785	825	870	910	950	990	1035	1075	1115	1155	1200	1240
4.9	780	820	865	905	950	995	1035	1080	1125	1165	1210	1255	1295
5.0	810	850	900	945	990	1040	1080	1130	1175	1220	1265	1310	1355
5.1	A 845	895	940	985	1035	1080	1130	1175	1220	1270	1315	1365	1410
5.2	880	930	980	1030	1080	1125	1175	1225	1275	1325	1370	1420	1470
5.3	920	970	1020	1070	1120	1175	1225	1275	1325	1375	1430	1480	1530
5.4	955	1005	1060	1115	1165	1220	1270	1325	1380	1430	1485	1535	1590
5.5	B 990	1045	1100	1155	1210	1265	1320	1375	1430	1485	1540	1595	1650
5.6	1030	1085	1140	1200	1255	1310	1370	1425	1480	1540	1600	1655	1710

Notes: Values below line A are above the specified average force.
Values below line B are above the specified individual force.

4.6.4 Penetration test. The penetration test shall be performed as follows to determine conformance with 3.5.3.

4.6.4.1 Apparatus. The test apparatus shall consist of standard head forms or test blocks as specified in 4.6.3.1(a) and a 16-ounce plumb bob (penetrator ball) with a steel point having an included angle of 35 ± 1° and a maximum point radius of 0.010 inch.

4.6.4.2 Procedure. Specimens shall be mounted as specified in 4.6.3.3.1 and placed on a substantially level concrete floor beneath the plumb bob. The plumb bob shall be dropped 10 feet to strike the shell within a 3 inch diameter circle, the center of which shall be the geometric center of the shell. The plumb bob shall not fall on any ridges or injection points. Test results shall be measured along the side of the point of the plumb bob to the nearest 1/32 inch and shall include the thickness of the shell.

4.6.4.3 Reporting. The depth of penetration shall be reported as the average of 3 specimens.

4.6.5 Flammability test. Method 2021 of Fed. Std. No. 406 shall be employed for the flammability test to determine conformance with 3.5.4. Three strips approximately 1/2 inch wide and 5 inches long shall be cut from the brim or sides of the crown, whichever is thinner, and shall be used in lieu of the 10 specimens required in method 2021.

4.6.5.1 Test conditions. Test specimens shall be subjected to the flammability test after subjection to the tests specified in 4.6.3 and 4.6.4.

4.6.5.2 Reporting. The burning rate reported shall be the average for the three specimens tested.

4.6.6 Water absorption test. Method 7031, immersion procedure "A" of Fed. Std. No. 406 shall be employed for the water absorption test to determine conformance with 3.5.5 with the following exceptions: Three whole helmet shells prepared in accordance with 4.6.1, when necessary, shall be used in lieu of the cut specimens, and conditioning in an oven at 120°F. for 4 hours and weighed before immerson in lieu of conditioning required by method 7031. Water absorption reported shall be the average of that determined for the three specimens.

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4.6.7 Headband clearance measurement. To determine conformance with 3.4.4, clearance between the headband and helmet shall be determined by measuring the circumference of the outside of the headband (at maximum size for adjustable headbands) and of the inside of the shell at the juncture of the dome and the visor and computing the diameter and clearance by the following formulas:

$$\text{Diameter} = \frac{\text{Circumference}}{\pi}$$

$$\text{Clearance} = \frac{D-d}{2}$$

D = inside diameter of shell.
d = outside diameter of headband (maximum size)

4.6.8 Laces or cords. The laces or cords when used for the crown strap assembly, 3.4.6, shall be tested for nylon content using any standard commercial test.

4.6.9 Weight. Helmets shall be weighed without winter liner, chinstrap, nape strap, and other accessories, on a suitable weighing balance to the nearest tenth (0.1) ounce to determine conformance with 3.5.6.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Level A. Each helmet shall be placed in a polyethylene bag fabricated from film at least 0.0015 inch in thickness. The bag shall be of sufficient length to enclose the helmet and permit the open end of the bag to be folded and tucked inside the helmet shell. Five wrapped helmets shall be stacked together by resting the crowns one within the other.

5.1.2 Level B. Unit packaging shall be in accordance with 5.1.1.

5.1.3 Level C. Unless otherwise specified in the contract or order, commercial packages are acceptable.

5.2 Packing.

5.2.1 Level A. Twenty helmets, packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each container shall be fitted with a taped liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-F-320. Each container shall be closed, waterproofed, and reinforced with strapping or tape banding in accordance with appendix of PPP-B-636.

5.2.2 Level B. Twenty helmets, packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Each shipping container shall be provided with a full height slotted partition made of double-faced corrugated fiberboard which shall divide the box into four equal sections. The container and inner packing shall be designed so that a stack of five packaged helmets will fit snugly in the partitioned section of the box. The inside of each container shall be fitted with a taped liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-F-320. Each container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.1 When specified (see 6.1), the shipping container shall be grade V3c or V3s fiberboard box fabricated in accordance with PPP-B-636, and closed in accordance with the appendix of the box specification.

5.2.3 Level C. Twenty helmets packaged as specified in 5.1 shall be packed in containers that will assure carrier acceptance and safe arrival at destination in compliance with the Uniform Freight Classification rules and the National Motor Freight Classification rules.

5.2.4 Standard pack (civil agencies). The standard pack for this commodity shall be as specified in 5.1.2 and 5.2.2.1.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking specified in the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military requirements. In addition to any special marking required by the contract or order, shipping container shall be marked in accordance with MIL-STD-129.

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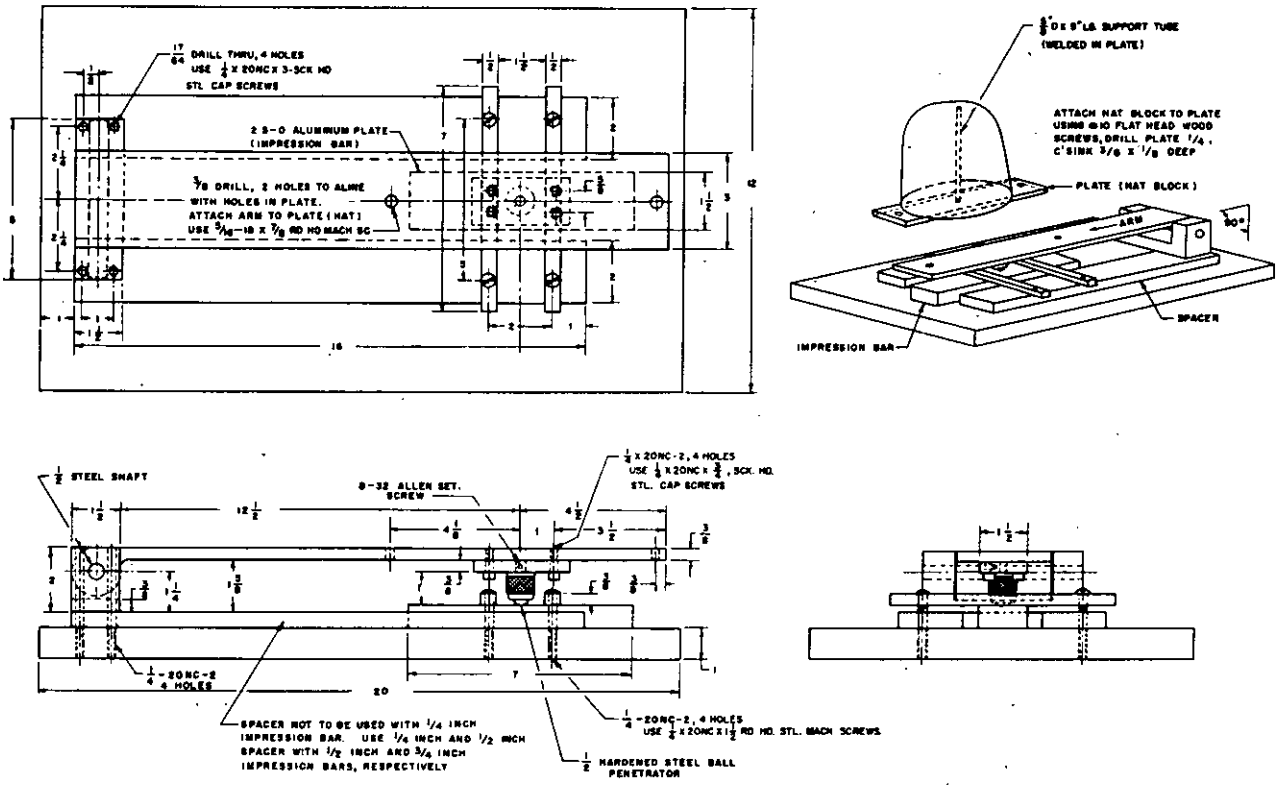


FIGURE 1. Brinell hardness penetrator assembly

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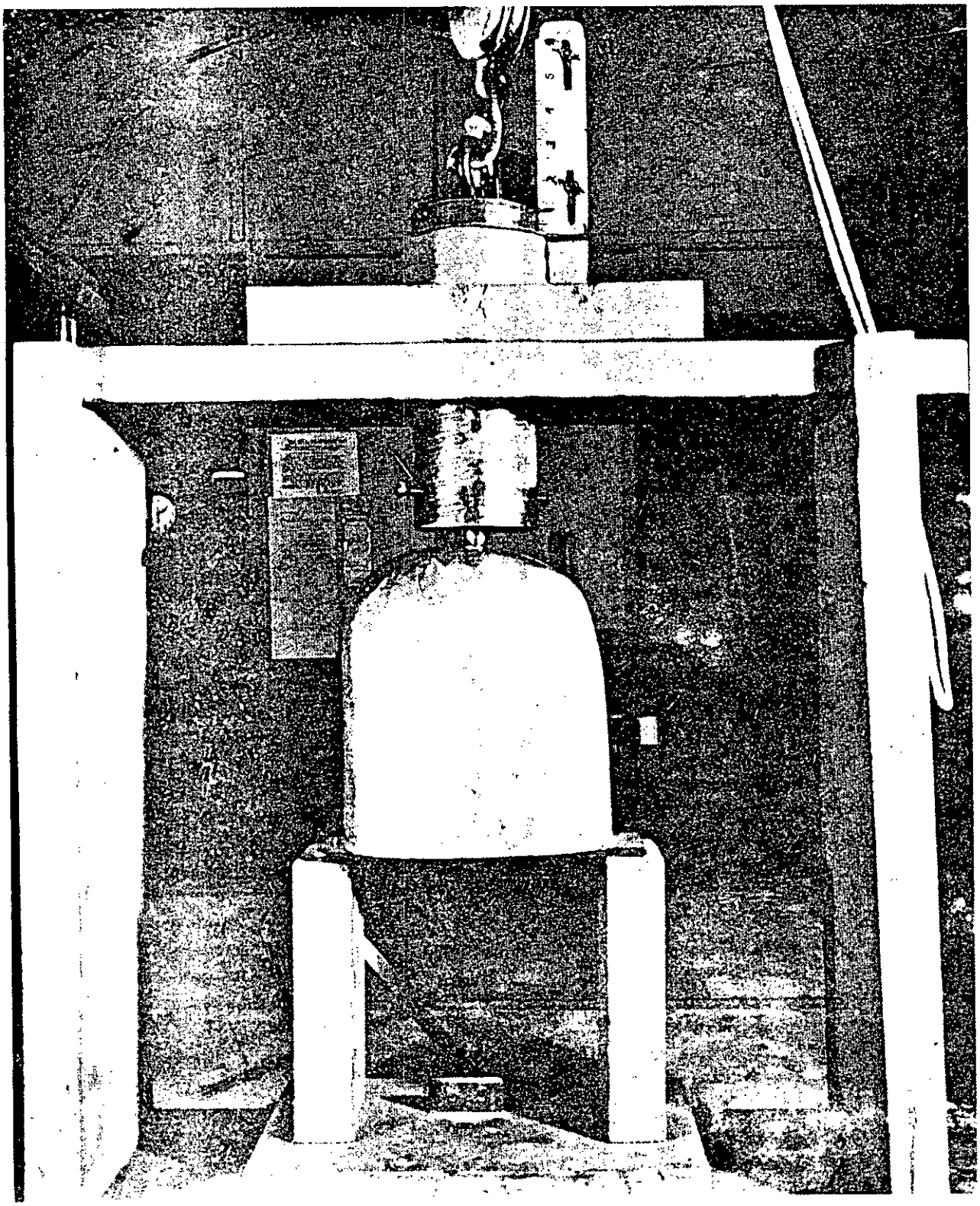


FIGURE 2. Brinell hardness penetrator

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6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2.1).
(c) If adjustable chinstrap and winter lining are required (see 3.2).
(d) When weather-resistant grade V3c or V3s fiberboard shipping containers are required for level B packing (see 5.2.2.1).
(e) Selection of applicable levels of packaging (see 5.1 and 5.2).

6.1.1 When headbands, sweatband, or crown strap assemblies for replacement purposes) are ordered, the manufacturer's type or model number of the helmet for which the replacement is ordered should be stated, or if possible, a representative helmet should be furnished in order that the manufacturer may have the necessary identifying information.

6.2 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion on the Federal Qualified Products List, whether or not such products have actually been so listed by that date.

6.2.1 The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Standardization Division, Federal Supply Service, General Services Administration, Washington, DC 20406.

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